



FEA Analysis and Redesign of Car Chassis

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MECE 4290U

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November 28th, 2025

Problem Definition

The objective of this project is to analyze the structural performance of a car chassis using Finite Element Methods (FEM). The chassis is required to have a total mass of 10,000 kg and a center of gravity positioned at (4.5m, 0m, 0.4m) from the front face. Structural steel was used with a Young's modulus of 210 GPa, Poisson's ratio of 0.3, and density of 7850 kg/m³.

The analysis includes global and local stress evaluation, bending stiffness, torsional stiffness, and mesh independence studies. A redesign of the chassis was also done with the objective of reducing chassis mass by at least 15% without significant loss of stiffness or structural performance.

Model Assumptions and Simplifications

Suspension Simplification

The suspension mounts were idealized using square patches created using the split tool in SolidWorks. These patches provide simplified and well-defined surfaces for applying forces and constraints.

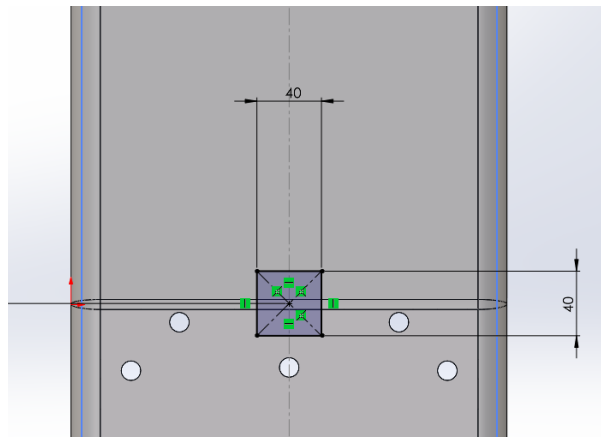


Figure 1: Suspension Mount Profile

The patches were centered 1 meter from the front face and 2 meters from the rearmost face, as specified in the instruction document.

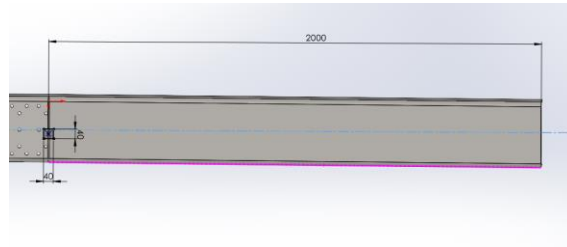


Figure 2: Rear Suspension Mount

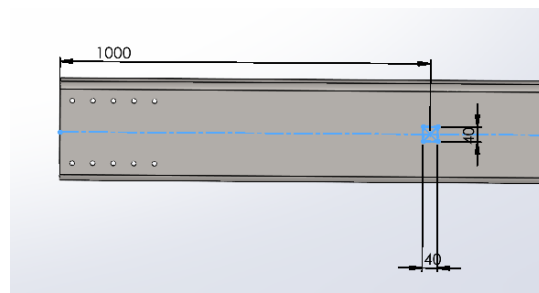


Figure 3: Front Suspension Mount

Connection Simplification

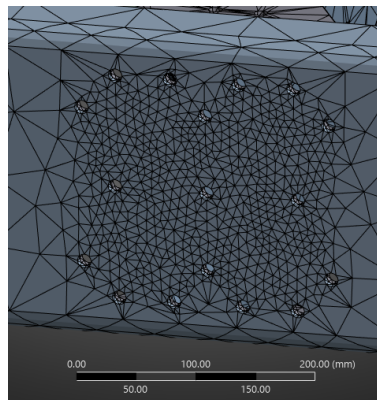


Figure 4: Fastener Holes Causing Adaptive Sizing to Create Many Small Elements

The numerous fastener holes in the chassis geometry were removed and replaced with bonded contact connections in ANSYS. This choice avoids excessive mesh refinement around the fastener holes which may contribute to meshing errors. It also provides an acceptable

approximation of structural continuity, as the large number of bolts in the chassis effectively creates near-rigid connections.

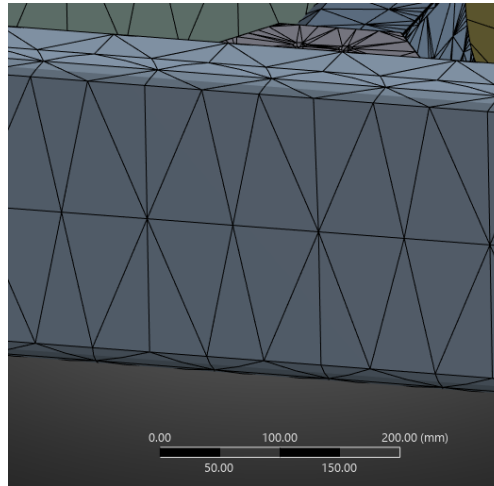


Figure 5: C-Beam with Fastener Holes Removed Has Significantly Less Elements

Center of Gravity

ANSYS does not allow direction reassignment of the chassis mass distribution, so a remote point mass was added to shift the center of gravity (COG) to a computed location and correct the total mass. After using the recommended material settings ($E = 210 \text{ GPa}$, $\nu = 0.3$, $\rho = 7850 \text{ kg/m}^3$), the total mass of the ladder was calculated to be 562.15 kg. A point mass was created with mass equal to $10000 \text{ kg} - 562.15 \text{ kg} = 9437.85 \text{ kg}$.

For the computation of the center of gravity (COG), the global coordinate systems origin started at the front crossmember of the ladder which matched the coordinate system from the instructions. From Ansys, the COG was found to be at -4.1802, $5.99\text{e-}008$, and $4.05\text{e-}004$. Knowing this and the required COG, the point mass would need to be placed at coordinates -4.518, 0, 0.423 from centroid calculations and the new COG was found to be -4.49, $0.233\text{e-}02$, 0.4 which matched the required COG.

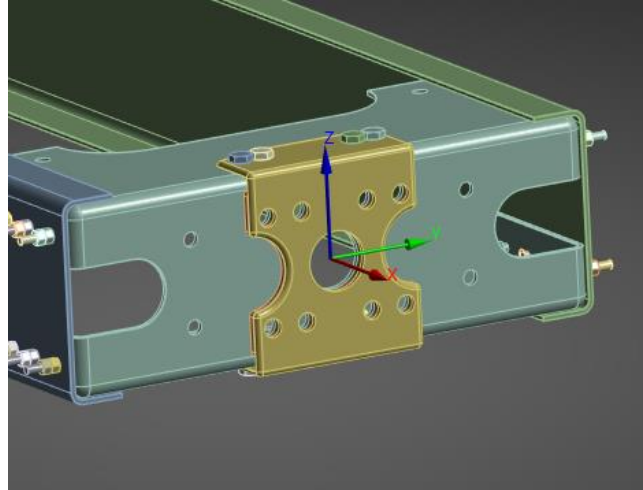


Figure 6: Global Coordinate System Starting at Front Face

Boundary Conditions and Meshing

Boundary Conditions

Boundary conditions were selected to reflect realistic test-rig behaviour for stress, bending, and torsional analyses.

Stress Analysis Boundary Conditions

For the stress analysis, each suspension point was constrained in a manner similar to a real vehicle suspension mount. These constraints prevented unrealistic rigid-body motion while allowing the appropriate structural deformation under the applied loads. These constraints are tabulated in the table below.

Suspension Location	Translational Movements	Rotational Movements
Front Right	Z constant	XYZ Constant
Front Left	Y and Z constant	XYZ Constant
Back Right	X and Z constant	XYZ Constant
Back Left	XYZ Constant	XYZ Constant

Table 1: DOF Constraints for Stress Analysis

Bending Stiffness Analysis Boundary Conditions

For the bending stiffness analysis, the rear suspension points were completely fixed in translation and rotation, representing the chassis being anchored to a rigid test fixture. The front suspension points were free to displace vertically, and an upward 2000 N load was applied at each location,

creating a total front-end load of 4000 N. The constraints and forces for this analysis are tabulated in the table below:

Suspension Location	Translational Movements	Rotational Movements	Force Applied
Front Right	XYZ Free	XYZ Free	2000 N in Z-direction
Front Left	XYZ Free	XYZ Free	2000 N in Z-direction
Back Right	XYZ Constant	XYZ Constant	0
Back Left	XYZ Constant	XYZ Constant	0

Table 2: DOF Constraints and Forces for Bending Stiffness Analysis

Torsion Stiffness Analysis Boundary Conditions

For the torsional stiffness analysis, the rear suspension point remained fully fixed while the front suspension points were subjected to equal and opposite vertical forces. A +ve 2000 N force was applied to the front right suspension mount, and a -ve 2000 N force was applied to the front left mount. This loading introduced a torque on the chassis which allows for analysis of the torsional stiffness. The constraints and forces for this analysis are tabulated in the table below:

Suspension Location	Translational Movements	Rotational Movements	Force Applied
Front Right	XYZ Free	XYZ Free	2000 N in Z-direction
Front Left	XYZ Free	XYZ Free	-2000 N in Z-direction
Back Right	XYZ Constant	XYZ Constant	0
Back Left	XYZ Constant	XYZ Constant	0

Table 3: DOF Constraints and Forces for Torsional Stiffness Analysis

Meshing

The mesh refinement strategy combined both global and local element sizing adjustments to achieve accurate and mesh-independent results. The initial global element size of 200mm provided a coarse representation suitable for early testing but it lacked the resolution needed for accurate analysis of curved and highly stressed features. Using a progressive refinement, the global element size was reduced by 20mm for the first 5 refinements, and then further by 30mm for the rest of the refinements. This refinement improved the accuracy of the stiffness and stress calculations.

Local mesh refinement was applied to regions with geometric complexity such as the crossmembers and towing jaw. These parts contained curved features and were generally smaller than the C-beams, so a smaller element size was required to accurately capture the behavior of these parts in the simulation. For these areas, a mesh size equal to one half of the global mesh size was used, and for the final test a local sizing of two thirds was chosen to reduce computational cost. The local element sizes were refined from 100mm down to 7.5mm in the final mesh. This approach allowed critical regions to be captured with a sufficient resolution while preventing unnecessary mesh density in less important areas, giving a good balance between accuracy and computational cost.

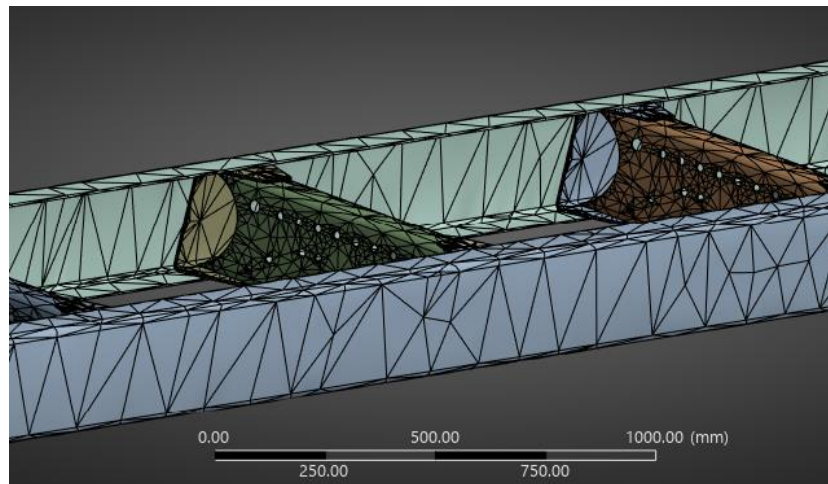


Figure 7: Mesh with 200mm Global Sizing on C-beams and 100mm Local Sizing on Crossmembers

Mesh Independence Studies

Mesh independence was evaluated using ten levels of progressive global and local mesh refinement. Each refinement level was used to compute global stress, local stress, bending stiffness, and torsional stiffness, which allowed the influence of mesh density on these parameters to be examined. The goal of the mesh independence studies was to identify the coarsest mesh that produced stable, mesh-independent results while avoiding unnecessary computational cost.

The global stress was first examined to determine whether peak stress values stabilized with mesh refinement. Both global and local stress metrics were evaluated to separate true structural behavior from artificial stress originating from boundary conditions and discontinuities.

Global Stress Convergence

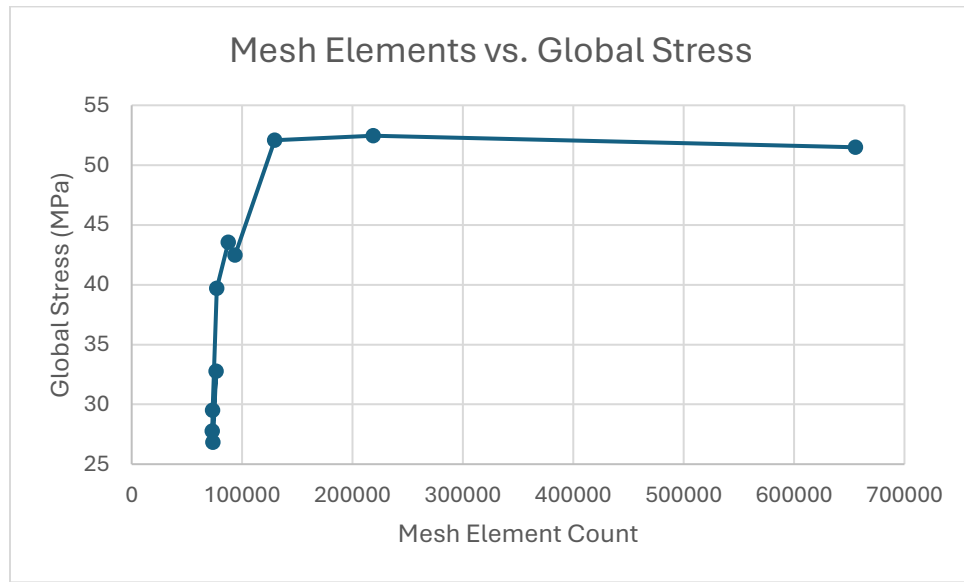


Figure 8: Mesh Elements vs. Global Stress Plot of Chassis

The plot of the mesh elements versus global stress shows that beyond roughly 150,000 elements, the maximum stress converged to a stable value of approximately 51-52 MPa, indicating that mesh independence was achieved for the global stress study. This also indicates that the fillet region is exhibiting stress riser behavior because it has a finite stress once the mesh is sufficiently refined.

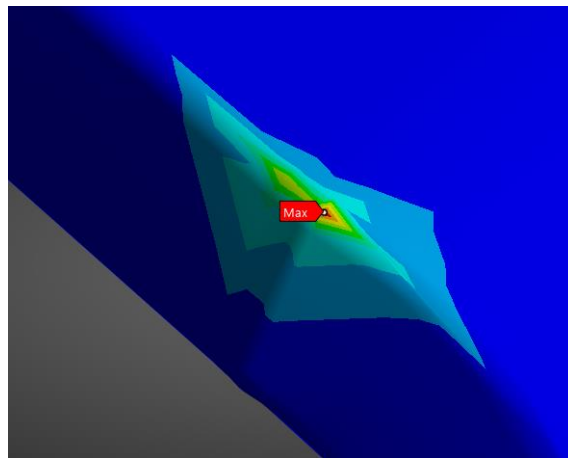


Figure 9: Max Stress Area on Fillet Edge

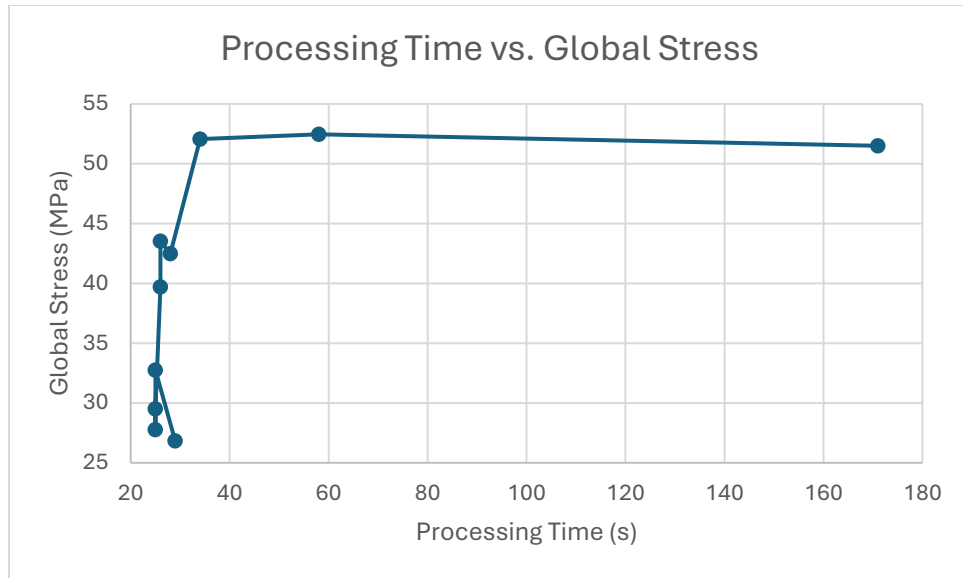


Figure 10: Processing Time vs. Global Stress Plot of Chassis

From the plot of processing time versus global stress, it can be seen that there are diminishing returns beyond 35 seconds of processing time, with the max global stress result converging at about 51-52 MPa. For a large amount of studies, the 30mm global sizing and 15mm local sizing would be preferable due to its balance between accuracy and computational speed.

Local Stress Patch Convergence

To verify that stress convergence was not influenced by high stress concentrations at sharp fillets, a patch along the top face of the right C-beam was used to measure stress and conduct a local stress convergence analysis.

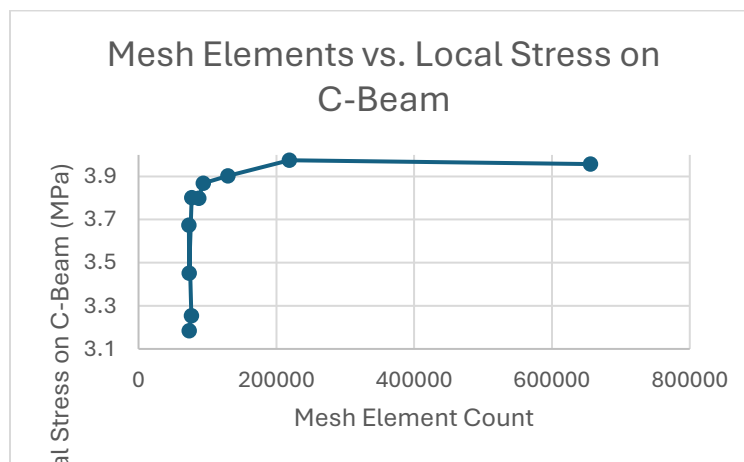


Figure 11: Plot of Mesh Element Count vs. Local Stress Patch

From the plot of the mesh elements versus local stress on the C-beam face, the local stress converged to a stable value of 4.0 MPa once the element count went beyond 150000 elements. This indicates that mesh independence was achieved for the local stress study.

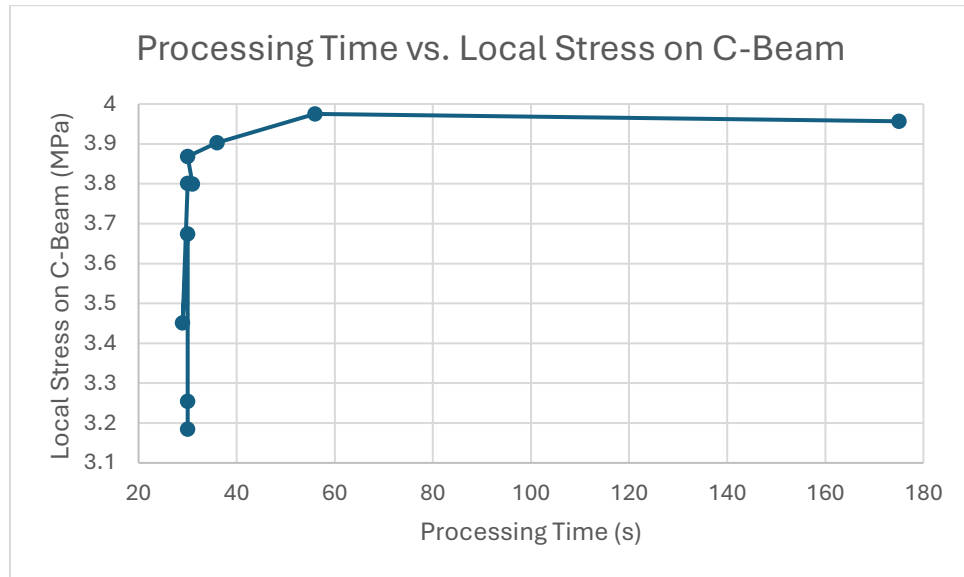


Figure 12: Plot of Processing Time vs Local Stress Patch

From the plot of the processing time versus local stress patch, it can be seen that the last study with 10mm global and 7.5mm local element sizing took significantly longer to process than the other studies and that there are diminishing returns past roughly 35 seconds of computation time. For running a large amount of simulations, it would be preferable to use the 30mm global and 15mm local sizing study as it offers accuracy and relatively fast processing times compared to the last study with the highest element count.

Bending Stiffness Convergence

After confirming the stress convergence, the bending stiffness was examined to determine whether deformation results were mesh independent. The results from the ten refinements are shown below:

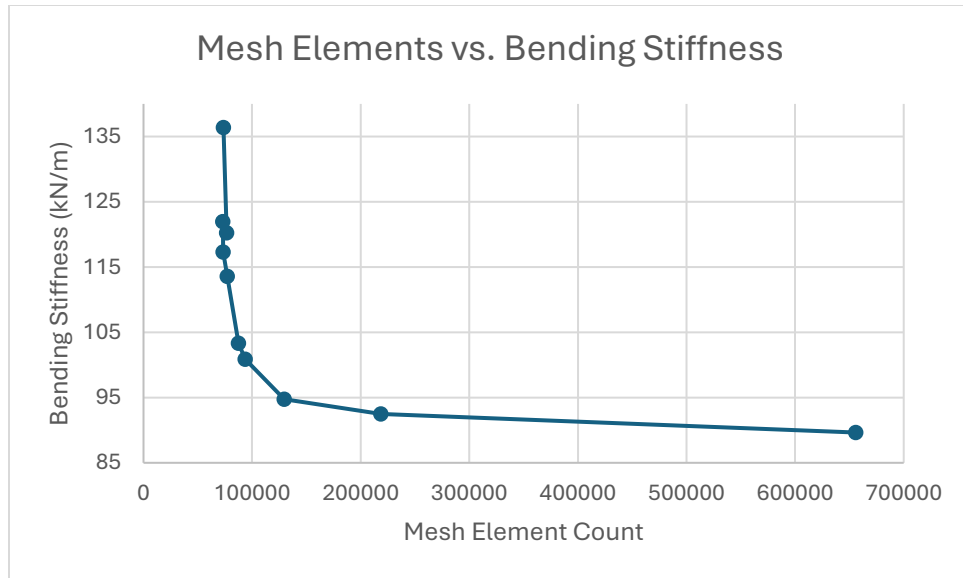


Figure 13: Plot of Mesh Element Count vs. Bending Stiffness of Chassis

From the plot of the mesh elements vs bending stiffness, the bending stiffness converges to a stable value of approximately 90 kN/m once the element count went beyond 200000. This indicates that mesh independence was achieved for the bending stiffness study.

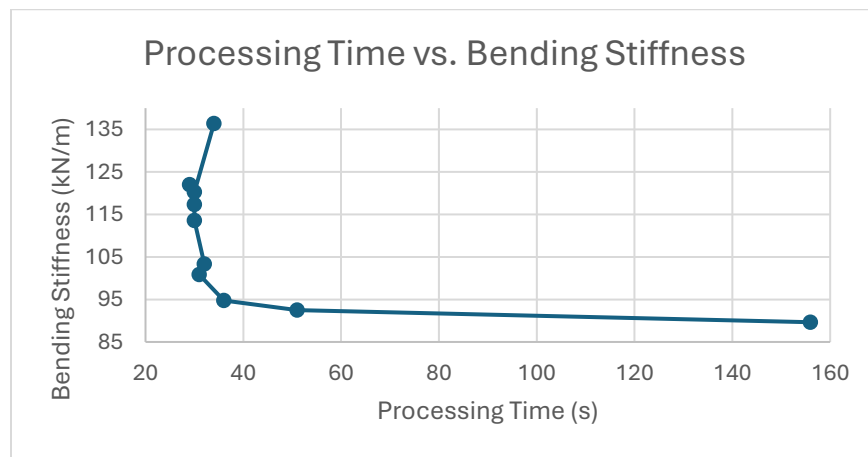


Figure 14: Plot of Processing Time vs. Bending Stiffness for Chassis Simulation

From the plot of the processing time vs. bending stiffness, the simulation time greatly increases from the 30mm global and 15mm local size to the 10mm and 7.5mm local size. It is also apparent that there are diminishing returns past roughly 40 seconds of processing time. If

running a large amount of simulations on the car chassis, the 30mm global and 15mm local sizing would be preferable due to the balance between accuracy and processing speed.

Torsional Stiffness Convergence

The convergence of torsional stiffness was evaluated using the measured rotation about the cylindrical axis between the two front suspension mounts. Similar to the bending stiffness results, the torsional stiffness showed a smooth convergence when the mesh density was increased.

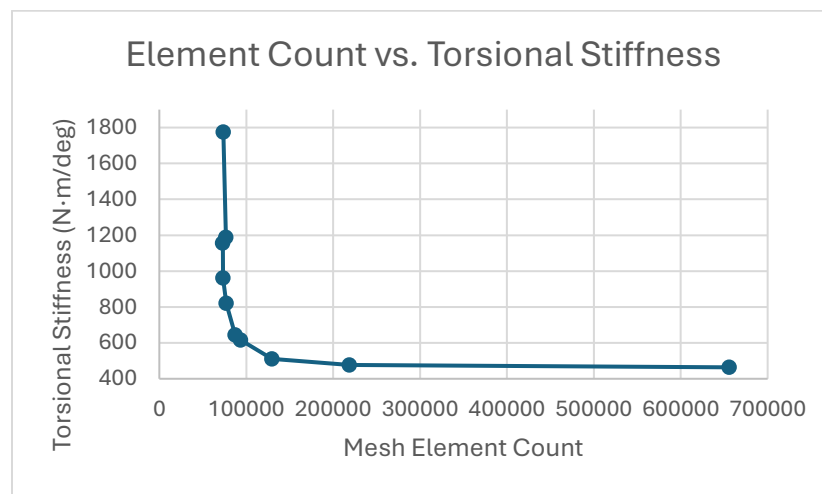


Figure 15: Plot of Element Count vs. Torsional Stiffness for Chassis

From the plot of the mesh elements vs torsional stiffness, the torsional stiffness converges to a stable value of approximately 490 N·m/deg once the element count went beyond 200000. This indicates that mesh independence was achieved for the torsional stiffness study.

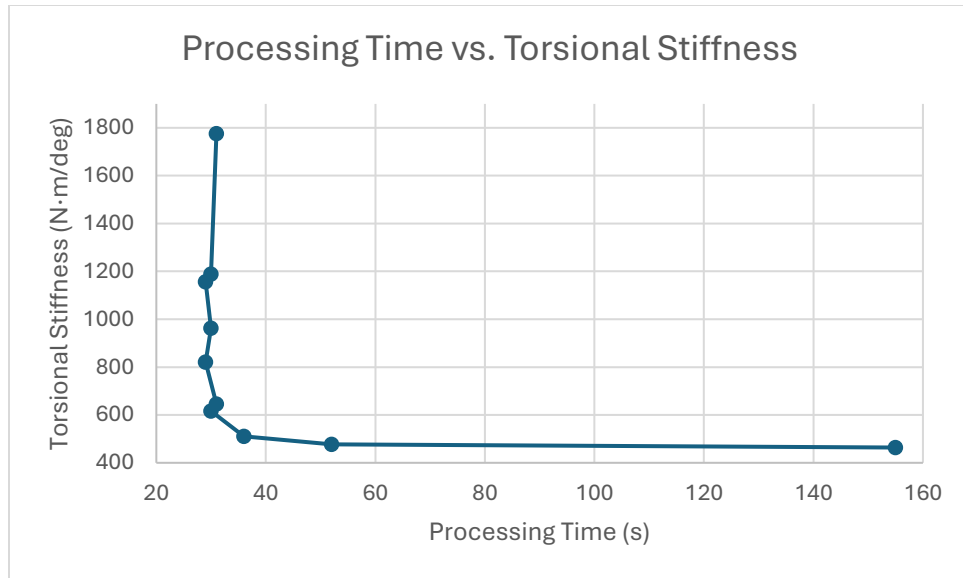


Figure 16: Plot of Processing Time vs. Torsional Stiffness

From the plot of processing time versus torsional stiffness, the plot also converges well to about 500 Nm/deg past roughly 35 seconds of processing time. The processing time increases dramatically when going from 30mm global and 15mm local sizing to 10mm and 7.5mm sizing, For a large amount of studies, the 50mm and 25mm sizing gives a very accurate result for a fraction of the processing time.

Analysis Results

Stress Analysis

Although the global stress contour plots displayed very high localized stress at the suspension attachment points, these peaks were the result of the square shape of the suspension mount causing high stress points at sharp corners and the fixed nature of the suspension mount imposed by the boundary conditions.

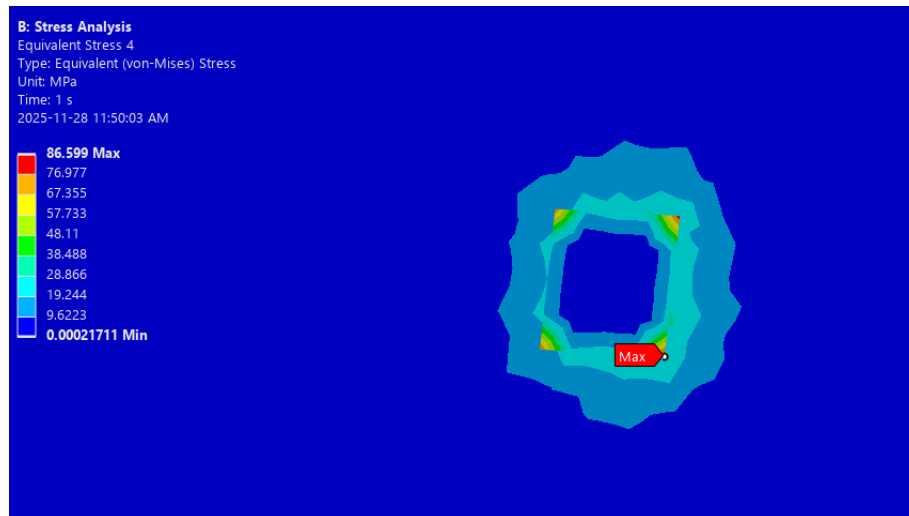


Figure 17: High Stress Area due to Suspension Mount

Using the scoping tool in ANSYS, the suspension mounts were removed, and a more accurate representation of the high stress zones could be found. The highest stress zone was found to be along the edge of the fillet of the crossmember near the back right suspension mount. There were around 10 MPa-20MPa across all of the crossmembers fillet edges but this point in particular had a stress of around 51 MPa which indicate that it is a critical region. This point also exhibits stress riser behavior as it has a finite value as evidenced by the mesh convergence plot.

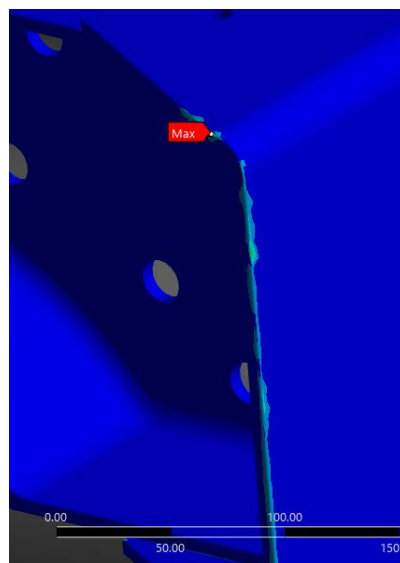


Figure 18: Max Stress Area in Chassis

Since the stress was so high at this point, it was difficult to give an overall assessment of the structural integrity as the whole chassis appeared to have negligible stress compared to those high stress areas as shown below:

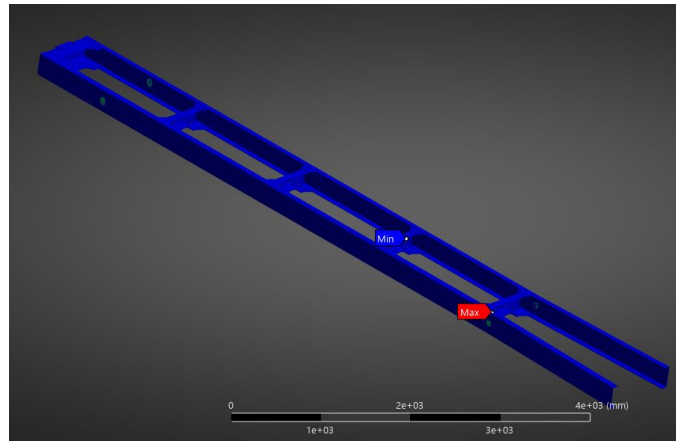


Figure 19: Contour Plot with Min and Max Stress

To fully understand the structural integrity of the chassis, the stress scale was limited to 4 MPa which gave a greater picture of the true stress distribution. The contour plot with the limited stress scale showed that the top and bottom areas of the C-beam midspan regions carried the majority of the structural load while the crossmembers experienced much lower stress.

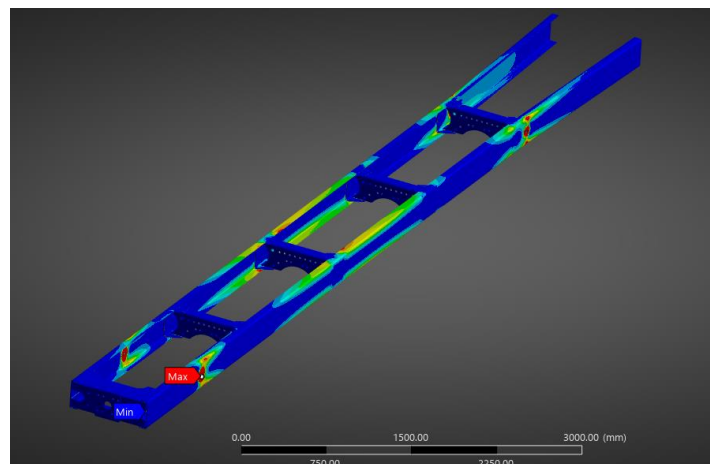


Figure 20: Stress Distribution on Chassis with 4 MPa Limit

Bending Stiffness Analysis

From the deformation contour plot, the most deformation was seen at the front of the chassis as expected.

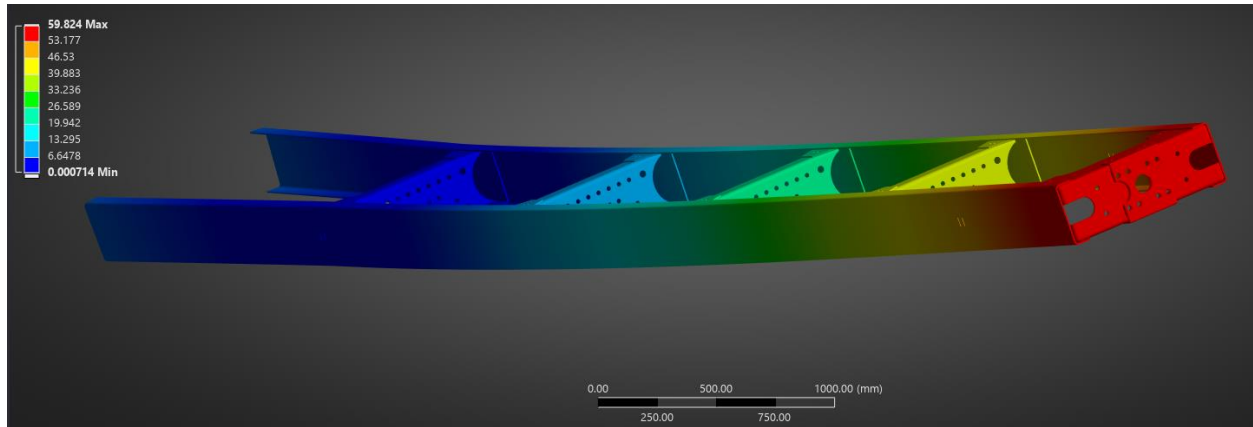


Figure 21: Deformation Contour Plot of Chassis under Bending

For the bending stiffness calculation, a deformation probe was placed on the front left suspension measure to measure the deformation in the z-axis from the given load of 4000 N on the chassis.

The equation used was $k = \frac{F_{total}}{d}$, with d being the measured deformation. This equation was used in all 10 studies which converged to a bending stiffness of roughly 90 kN/m.

Torsional Stiffness Analysis

For the torsional stiffness analysis, a cylindrical coordinate system was made between the two front suspension points to measure the distance it rotates. These values were recorded for all 10 refinement levels and were later used to calculate the torsional stiffness.

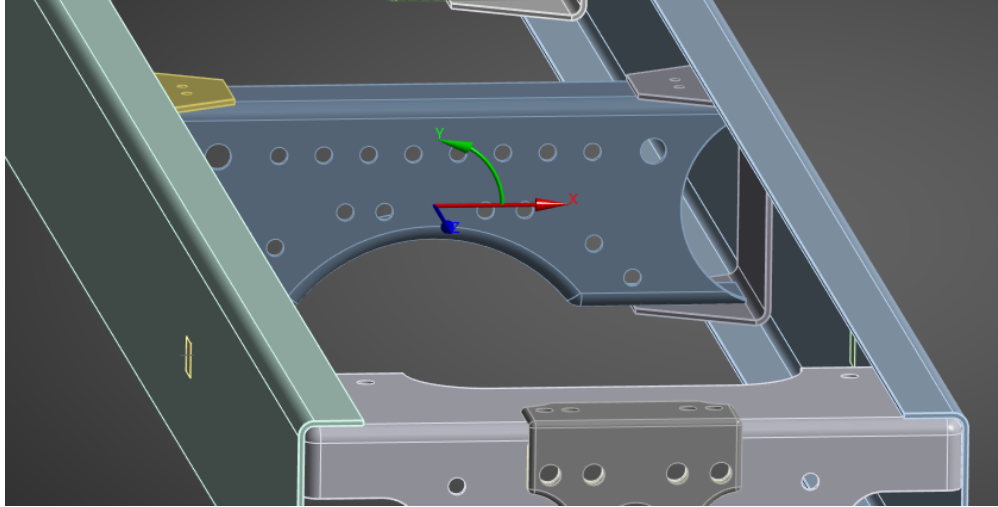


Figure 22: Cylindrical axis between the front suspension points of chassis

The contour plot of the torsional deformation showed the highest deformation about the front right and left tips of the chassis, with little rotation in the center of the crossbeams.

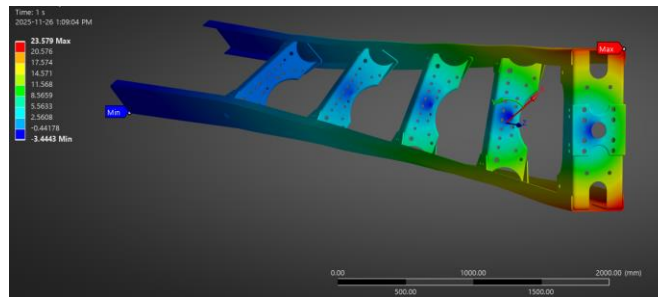


Figure 23: Contour Plot of Torsional Deformation of Car Chassis

The equation used to find the torsional stiffness was $k_t = \frac{Torque(\frac{180}{\pi})}{(\frac{d_{measured}}{radius})} = \frac{2(F)(radius)(\frac{180}{\pi})}{(\frac{d_{measured}}{radius})}$, with the force used being 2000 N and the radius measured at 0.385m, which is half the distance between the two front suspension mounts. This torsional stiffness was calculated for all 10 studies and converged to a torsional stiffness of about 490 N-m/deg.

Design Modification for Weight Reduction

The calculated mass before any modifications were made in SolidWorks was found to be 540.99 kg. With a goal of 15% weight reduction, this gives a maximum weight of 460 kg for the redesigned chassis, with the mass being relative.

Modified stress contour plots were created by limiting the maximum stress in the study. By doing this, a higher level analysis of the car chassis could be conducted. Looking at modified stress contour plots of the initial stress, bending stiffness, and torsional stiffness analysis, several areas were found to be low in stress in all studies. These areas were then modified by cutting out material in circular shapes to avoid high stress concentrations.

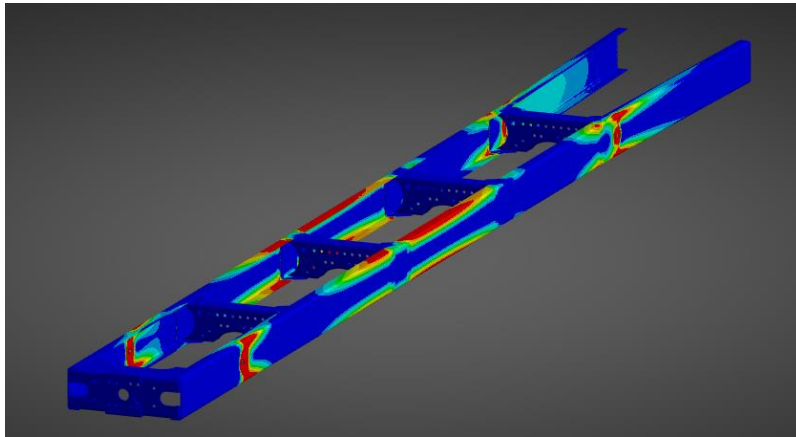


Figure 24: Modified Stress Analysis

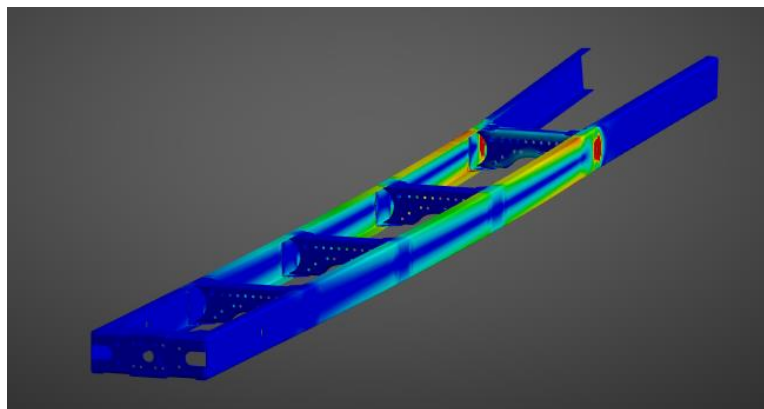


Figure 25: Modified Bending Stiffness Analysis for Stress

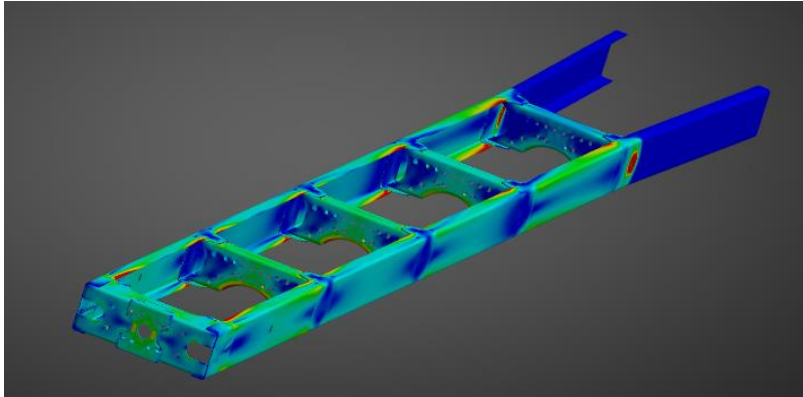


Figure 26: Modified Torsional Stiffness Analysis for Stress

Even though the chassis can handle much more stress, and material could be cut/thinned all throughout the chassis, the objective of this section was to make the chassis lighter with no reduction in strength or stiffness. The approach taken for this was to cross-reference all 3 stress studies shown above and find areas where all 3 contour plots have low stress. In those areas, the material can be removed safely.

The primary region where the weight reduction was achieved was in the C-beams. They show low stress areas all along the rear end and the midsection of the C-beams. Cuts were made across those regions, and additional cuts were made to the top and bottom faces of the C-beam near it's rear as shown below:

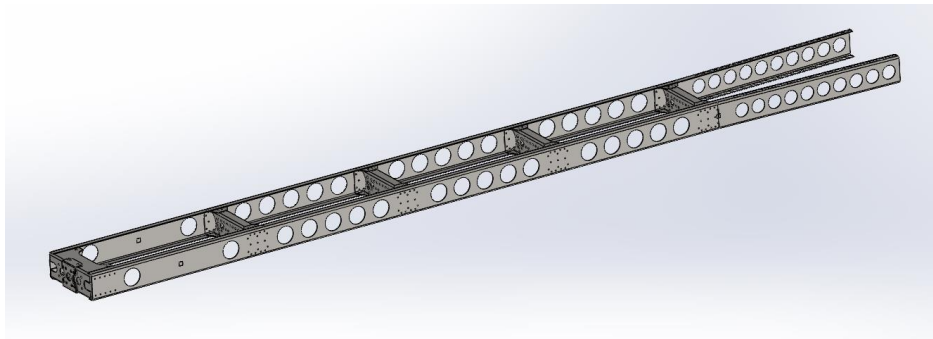


Figure 27: Extrusions along c beam which avoid high stress areas

Another low stress area was to the left and right of the centered semicircle in the cross-members. All three studies show that there was a low stress area at that point, so material was removed there.

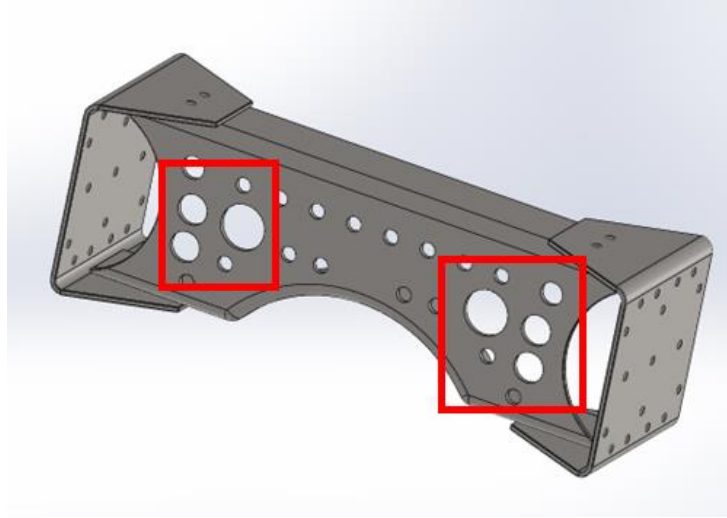


Figure 28: Extrusions to left and right of semicircle

The last area where all three studies showed a low stress point was the left and right of the circle in the middle of the towing jaw assembly. Material was also removed in that region.

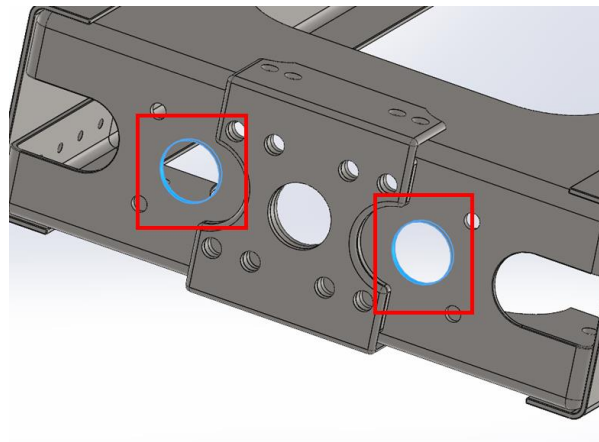


Figure 29: Extrusions to the Left and Right of Towing Jaw

All the material removed was done so via a circular cut, as a cut with sharp corners such as a triangle or square would impose high stress concentrations at the sharp edges. The circular cut would have a smooth, even stress concentration.

The weight reduction which was achieved was calculated to be 15.8%, with the new weight of the chassis at 467 kg.

The 10mm global and 7.5mm local element mesh sizing were used to evaluate the performance of the redesigned chassis as it only took 2 minutes to process and had the highest quality mesh.

Stress Analysis for Redesign

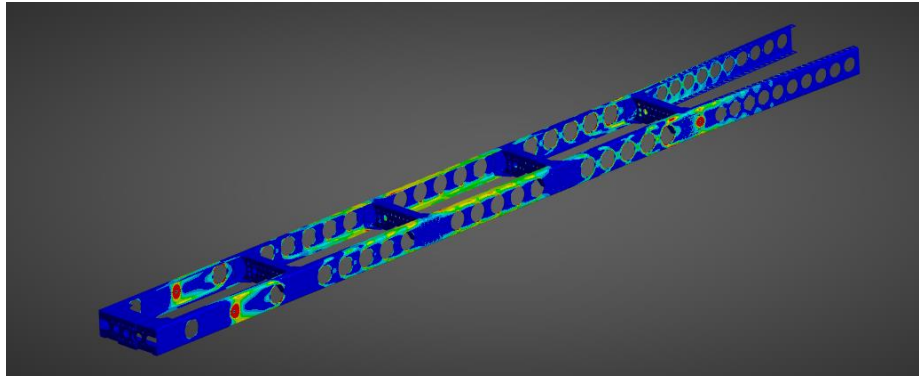


Figure 30: Redesigned Mesh Stress Analysis

From the analysis, there were max stresses around 4 MPa on the middle area of the C-beams, with the stresses near the suspension mounts higher of course. These stresses are well within the range that structural steel can handle.

Bending Stiffness Analysis for Redesign

The bending stiffness for the redesigned chassis was found to be 82.497 kN/m versus the original 89.642 kN/m bending stiffness. This represents an 8% decrease in bending stiffness which is negligible and can be easily increased if needed by filling certain extrusions at critical areas.

Torsion Stiffness Analysis for Redesign

The torsional stiffness for the redesigned chassis was calculated to be 436.4 N-m/deg versus the original 463.8 N-m/deg torsional stiffness. This decrease represents a 6% decrease in torsional stiffness which is also negligible and can also be increased like the bending stiffness, if necessary.

Validation and Discussion

Using beam equations, the 2 C-beams were approximated to a single cantilever beam with 4000 N at the tip, which closely resembles the simulation setup. The length in the hand calculation is the length between the front and rear suspension points.

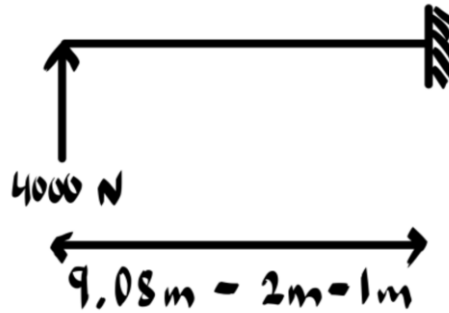


Figure 31: Free Body Diagram of the Bending Stiffness Setup

Using SolidWorks, the second moment of area of the C-beam was found to be $2.636 \times 10^{-5} \text{ m}^4$ about the x-axis which is where the bending occurred in the simulation. Using the beam equation for deflection of a cantilever beam, $v_{max} = \frac{PL^3}{3EI}$, the maximum deflection was calculated to be

$$v_{max} = \frac{(4000 \text{ N})(6.08 \text{ m})^3}{3(210 \times 10^9)(2.636 \times 10^{-5} \text{ m}^4)} = 0.0541 \text{ m}.$$

Using the stiffness equation $k = \frac{F}{v_{max}}$, the stiffness was calculated to be 73.9 kN/m. This is close to the results found in the simulations, being 89.6 kN/m for the finest mesh study.

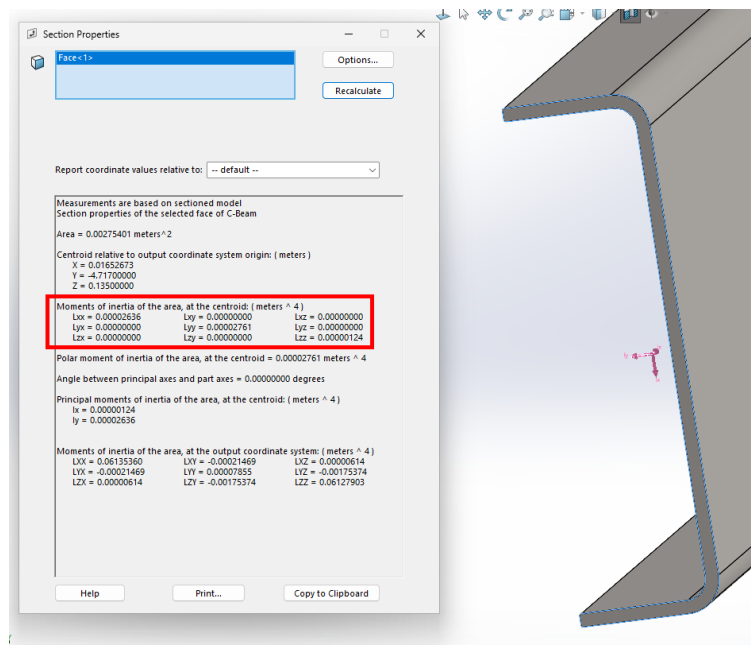


Figure 32: Second Moment of Area Calculated from SolidWorks

The boundary conditions made a significant impact on the stress analysis due to the high stress concentrations near the suspension. These high stresses shrouded the max stress value and alternative methods such as limiting the stress and hiding the support were needed to find more accurate stress results for the chassis.

The material assumption for structural steel proves to be a good material for this design, as none of the stresses got close to the yield stress of the structural steel with the maximum stress giving a factor of safety of 4.81.

Conclusions and Recommendations

The finite element analysis demonstrated that the chassis possesses good structural performance under stress, bending, and torsional loading. Mesh-independent behavior was achieved for all critical analysis studies, ensuring that the results are reliable. The baseline chassis exhibited a bending stiffness of approximately 90 kN/m and a torsional stiffness of about 490 N-m/deg, both consistent with expectations for a ladder-type structure. Peak stresses were localized at geometric fillets and remained well below the yield strength of structural steel.

Material removal performed during the redesign successfully reduced the chassis mass by 15.8% while maintaining bending and torsional stiffness within 6–8% of the original design. This demonstrates that significant weight reduction can be achieved without compromising structural integrity.

For further refinement, local reinforcements such as larger fillets or local thickness increases can be applied at critical stress-riser locations. Additional optimization could be achieved by modifying crossmember geometry to improve torsional stiffness while keeping weight low. The final design meets the performance objectives and represents a structurally efficient chassis configuration.

Appendix A – Calculated Values and Point Mass Location

Details of "Point Mass"	
Scope	
Scoping Method	Remote Point
Applied By	Remote Attachment
Remote Points	Remote Point
Coordinate System	Global Coordinate S...
X Coordinate	-4518.4 mm
Y Coordinate	0. mm
Z Coordinate	423. mm
Definition	
Mass	9455.5 kg
Mass Moment of Inertia X	0. kg.mm ²
Mass Moment of Inertia Y	0. kg.mm ²
Mass Moment of Inertia Z	0. kg.mm ²
Element APDL Name	
Suppressed	No

Details of Point Mass to Adjust COG Location

```
TOTAL MASS = 9.9972
The mass principal axes coincide with the global Cartesian axes
CENTER OF MASS (X,Y,Z)= -4499.8      0.23321E-02      400.10
```

ANSYS COG and Total Mass for Chassis

Mass = 540.99 kilograms

Original Mass of Chassis in SolidWorks